Yuan

Network Engineer | Infrastructure | Security Platforms | AI Automation

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Summary

Student, network technology enthusiast, and network engineer from Taiwan with experience in Network Engineering, Network Infrastructure, BGP, Routing and Switching, Linux System Operations, Security Training Platforms, AI Agent development, n8n Automation, and AI Application Development. Currently Founder at NCSE Network, Network Engineer at 諾科雲有限公司, and a student in the Department of Computer Science and Information Engineering at National Ilan University. Open to Network Engineer, Infrastructure Engineer, SRE, Security Platform Engineer, AI Automation Engineer, and Forward Deployed Engineer roles or collaborations.

Skills

- Network Engineering: BGP, network architecture, routing and switching, Cisco Routing and Switching, Juniper Routing and Switching, VyOS, peering, DNS, IPv6, Looking Glass, network troubleshooting, network automation.
- Network Operations: SNMP monitoring, Linux server administration, service operations, data processing, data analysis systems, operations tooling.
- Systems and Virtualization: Debian GNU/Linux, Ubuntu, Proxmox VE, HomeLab, Docker, containerization.
- Security Platforms: Web Security, CTFd operations, CTF platform operations, vulnerable machine maintenance, security training platform operations, containerized challenge deployment.
- AI Applications and Automation: AI agent development, n8n workflow automation, RAG, document understanding, Moodle integration, AI application development.
- Web Tools and Integrations: PHP, C++, NodeJS, API development, Discord bot development, Google Sheets API, WHMCS plugin development, payment gateway integration.
- Community and Education: technical education, community operations, event operations, information security teaching, curriculum support.

Work Experience

NCSE Network - Founder
2022/06 - Present

- Built and maintained NCSE Network compute, network, and service infrastructure from scratch.
- Worked on network planning, system operations, service operations, and internal tooling improvements.
- Operated infrastructure and services related to network engineering, automation, monitoring, Looking Glass, and public network tools.

諾科雲有限公司 - Network Engineer
2025/10 - Present

- Support network service projects, internal management workflows, data integration, and analysis systems.
- Assist with network architecture improvements, network troubleshooting, and operations tooling.
- Work across infrastructure, network operations, process automation, and operations workflow tooling.

美商西雲有限公司在臺辦事處 - Network Engineer
2025/05 - 2025/10

- Supported network service projects, internal management, data integration, and analysis system development.
- Assisted with network architecture improvement and network troubleshooting.

Ministry of Education Cybersecurity Talent Development Project, Subproject 3 - Part-time Assistant
2024/10 - 2025/05

- Assisted with security training platform administration.
- Maintained vulnerable machines and CTFd platform operations.
- Deployed CTF challenge services using containerized workflows.

Projects

LearnAI - Multi-user AI Learning Assistant Platform
GitHub: https://github.com/tw-yuan/niu-code-1142-project

- Built a multi-user AI learning assistant focused on course data, document understanding, AI-assisted Q&A workflows, and learning-context integration.
- Planned product workflows, integrated frontend and backend features, connected AI capabilities, and handled data-processing flows.
- Technologies and domains: AI application, RAG, document processing, web application.

NIU AI Buffet - External AI Tooling for National Ilan University Moodle Workflows
GitHub: https://github.com/tw-yuan/niu-ai-buffet

- Built tooling that allows National Ilan University Moodle AI capabilities to be used from external workflows.
- Designed external usage flows and integrated Moodle context, tool interfaces, and AI access flows.
- Technologies and domains: Moodle integration, AI application, automation tooling.

bgpq3 Web API
GitHub: https://github.com/tw-yuan/bgpq3-web-api

- Wrapped the bgpq3 network tool as a Web API for prefix-list and routing automation workflows.
- Designed an API interface so routing and prefix-list generation can be reused by other systems.
- Technologies and domains: BGP tooling, Web API, network automation.

NetToolsBot - Discord Bot for Network Diagnostics
GitHub: https://github.com/tw-yuan/NodeJS-NetToolsBot

- Built a Discord bot for network testing and diagnostics in chat-based workflows.
- Designed bot commands and network testing flows to reduce the friction of common network checks.
- Technologies and domains: NodeJS, Discord Bot, network diagnostics.

Additional Projects

- Code Noter - code sharing platform: https://github.com/tw-yuan/apcsc-code-noter
- Fontly - real-time font preview tool: https://github.com/tw-yuan/Fontly
- Pikachu Volleyball clone using C++ and Easy2D: https://github.com/tw-yuan/niu-pikapika
- SITCON relationship graph and relationship management system: https://github.com/tw-yuan/sitcon-relationship
- Discord x Google Sheets bookkeeping bot: https://github.com/tw-yuan/money-record-bot
- Convoy Panel WHMCS plugin: https://github.com/tw-yuan/convoy-whmcs
- SMSE WHMCS payment gateway integration: https://github.com/tw-yuan/SMSE-Whmcs
- OnePaid WHMCS payment gateway integration: https://github.com/tw-yuan/onepaid-whmcs
- Personal HomeLab for network, virtualization, system operations, and service deployment.

Education

National Ilan University - Department of Computer Science and Information Engineering
2024 - Present

Awards

- TWNOG Talent Development Program - 2026 Fellow, 2026.
- National Ilan University 2025 CTF Security Attack and Defense Capture-the-Flag Competition - Champion, 2025.
- 2023 T-Cat Cup National Cybersecurity Basic Practical Skills Competition - Excellent Award, 2023.
- Ministry of Education Cybersecurity Talent Development Project - MyFirstCTF 2022, participant.
- Ministry of Education Cybersecurity Talent Development Project - AIS3 Pre-Exam 2023, participant.
- Ministry of Education Cybersecurity Talent Development Project - AIS3 EOF 2023, participant.
- Ministry of Education Cybersecurity Talent Development Project - AIS3 Pre-Exam 2024, participant.
- Ministry of Education Cybersecurity Talent Development Project - AIS3 EOF 2024, participant.
- 9th National Cheng Kung University Summer High School Programming Invitational, participant.

Training

- AIS3 Advanced Training, 10th cohort - ongoing.
- Ministry of Education Cybersecurity Talent Development Project - AIS3 Junior 2026 - teaching assistant.
- Ministry of Education Cybersecurity Talent Development Project - AIS3 TOUR One-Day Cybersecurity Experience Camp 2026 North Taiwan - teaching assistant.
- Ministry of Education Cybersecurity Talent Development Project - AIS3 TOUR One-Day Cybersecurity Experience Camp 2026 Central Taiwan - teaching assistant.

- APNIC - TWNOG 7.0 IPv6 Workshop - completed.
- Ministry of Digital Affairs DNS Public Policy Workshop - participant.
- AIS3 Advanced Training, 9th cohort - completed.
- Taiwan Holy High Advanced Training, 8th cohort - completed.
- Ministry of Education Cybersecurity Talent Development Project - AIS3 Juniors 2024 - teaching assistant.
- SecurityFocus Online 2023 - participant.
- Ministry of Education Cybersecurity Talent Development Project - AIS3 2023 - participant.
- National Yang Ming Chiao Tung University Talent Development Program, Phase 15 - Artificial Intelligence and Deep Learning - participant.
- Ministry of Education Cybersecurity Talent Development Project - AIS3 Juniors 2022 - participant.
- Ministry of Education Cybersecurity Talent Development Project - AIS3 2022 - participant.
- Ministry of Education Cybersecurity Talent Development Project - 2022 Yunlin-Chiayi Cybersecurity Experience Camp - participant.
- Ministry of Education Cybersecurity Talent Development Project - 2021 Central Taiwan Cybersecurity Experience Camp - participant.

Speaking Experience

- COSCUP 2025 - HomeLab from networking: introductory VyOS tutorial for Linux users.
- TANET 2025 - eduroam, TANetRoaming, and international trend comparison.
- Ministry of Education Cybersecurity Talent Development Project - AIS3 East Taiwan Cybersecurity Experience Camp 2026 - Web Security.
- SITCON 2026 Lightning Talk - Liberal arts, SITCON, and AI Agent.
- Ministry of Education Cybersecurity Talent Development Project - AIS3 TOUR One-Day Cybersecurity Experience Camp 2026 East Taiwan - Web Security.
- Ministry of Education Cybersecurity Talent Development Project - AIS3 TOUR One-Day Cybersecurity Experience Camp 2026 South Taiwan - AI Security.
- Ministry of Education Cross-Disciplinary ICT Security Talent Development Program 2025 National Cybersecurity Club and Community Cadre Camp - community operations experience sharing.
- SCINT 2025 Basic Skills Course - network debugging methods.
- SCINT 2025 Basic Skills Course - firewall concepts and hands-on practice.
- One-Day Information Experience Camp 2024 Yunlin - network operations and governance.
- One-Day Information Experience Camp 2024 Tainan - network security and practice.
- SCINT 2024 Four-School Joint Course - network governance and operations.
- CISCON 2026 Community Month - network operations and network governance.
- High school club courses covering information security, programming, network basics, network governance, and network security practice.

Community Leadership

- SCINT North Taiwan Student Information Community - co-founder and security project lead.
- SITCON HoC One Hour of Code 2025 - nationwide coordination lead.
- TANET 2025 - paper author.
- THJCC Taiwan High School Joint Cybersecurity Competition - first edition network administrator, second edition lead coordinator, third edition network administrator.
- SCINT Joint Programming Contest, first edition - lead coordinator.
- SITCON 2027 - administration group lead and editorial group member.
- SITCON Camp 2026 - editorial group deputy lead.
- SITCON 2026 - editorial group deputy lead.
- SITCON 2024 - program group member and venue operations mobile team lead.
- SITCON Hackathon 2024 - information systems team member.
- SITCON Camp 2023, 2024, 2025 - information systems or editorial team member.
- COSCUP 2023 - staff.
- g0v Summit 2024 - staff.
- HITCON 2023 - staff.
- CYBERSEC Taiwan 2022 - 2026 - attendee.
- TWNOG 5.0, 6.0, and 7.0 - attendee.
- JANOG57 - attendee.
- ARIN 53 Public Policy and Members Meetings - participant.